today's (cloud) computers.[3] We model all risk and rewards through these statistical methods. Our notion of risk today is extremely broad—it is not just the likelihood of a set of returns. We think of risk today in terms of probability functions encompassing many different kinds of events (or even parallel universes, if you are a theoretical physicist).[4] We even think of risk as the probability of sets of probabilities changing over time, as I explain below.

Risk in financial economics is inherently a subjective concept. We do not have controlled laboratory conditions in finance and must estimate the distribution of asset returns. The procedures used to estimate probabilities and the model that is behind the estimation impart subjectivity. Nevertheless, we refer to direct estimates of the behavior of returns from data as being *real world measures* or *objective measures*. In contrast, *subjective measures* are probabilities that an asset owner believes in and that differ from real-world probabilities. A coin flip, for example, has a 50% objective probability of landing heads, but an investor might believe the coin is biased and assign a subjective probability of 60% to the coin coming up heads. Probabilities of returns are harder to estimate than probabilities of coin flips because there are an infinite number of return outcomes ranging from losing all your money (a return of –100%) to becoming a bazillionaire (a return of nearly infinity), and everything in between. (If you invest with leverage, your return may even be lower than –100%.)

In Panel A of Figure 2.2, I estimate objective probabilities for the volatility strategy at the monthly frequency from April 1989 to December 2011. The top half presents a histogram of returns and the bottom half graphs the strategy's *probability density function* (or *probability distribution function*, or just *pdf*). In both graphs, the *x*-axis lists the returns (what can happen), and the *y*-axis lists the probabilities of the return outcomes (how often they happen). The huge occasional loss that occurs in volatility investing, which happened during 2008–2009, imparts a very long left-hand tail to its probability distribution. The very long left-hand tail is the statistical representation of picking up the nickels and dimes right before the steamroller crushes you.

The estimated probability density function of the volatility strategy has a much thinner body and longer tail than a fitted normal distribution. We call these distributions *leptokurtic*, which is Greek for "thin hump," since they have much more slender distributions at the center than a normal distribution. When your distribution has a more slender body, it also has longer tails than a normal distribution. Figure 2.2, Panel B, repeats the same exercise for the S&P 500 for comparison. Equity returns over this time period are also leptokurtic, but much less so than the volatility strategy. While there is a noticeable left-hand tail for equities, the normal distribution is not a bad approximation. (While it is a much closer

[3] The 2008–2009 financial crisis shows that we remain far from mastering systemic risk.

[4] This is measure theory. Even the foundation of Brownian motion, which we use in option pricing and dynamic portfolio choice (see chapter 4), rests on probabilities specified over function spaces (called "functionals").